

9.1 (a) Consider a rotating charge distribution, $\rho(\vec{x}, t) = \rho(r, \theta, \phi - \omega t)$. The charge distribution can be expanded with spherical harmonics, $\rho(\vec{x}, t) = \sum_{l=0}^{\infty} \sum_{m=-l}^l \rho_{lm}(r, t) Y_{lm}(\theta, \phi)$, where

$$\begin{aligned} \rho_{lm}(r, t) &= \int_{-1}^1 d(\cos\theta) \int_0^{2\pi} d\phi \rho(r, \theta, \phi - \omega t) Y_{lm}^*(\theta, \phi) \\ &= \int_{-1}^1 d(\cos\theta) \int_0^{2\pi} d\phi \rho(r, \theta, \phi - \omega t) \sqrt{\frac{4\pi}{2l+1}} \frac{(l-m)!}{(l+m)!} P_l^m(\cos\theta) e^{-im\phi} \end{aligned}$$

After a change of variable $\phi \rightarrow \phi + \omega t$, the time dependence can be factored out,

$$\rho_{lm}(r, t) = \left[\int_{-1}^1 d(\cos\theta) \int_0^{2\pi} d\phi \rho(r, \theta, \phi) \sqrt{\frac{4\pi}{2l+1}} \frac{(l-m)!}{(l+m)!} P_l^m(\cos\theta) e^{-im\phi} \right] e^{-im\omega t} = \rho_{lm}(r) e^{-im\omega t}$$

Now, the charge distribution can be written as

$$\begin{aligned} \rho(\vec{x}, t) &= \sum_{l=0}^{\infty} \sum_{m=-l}^l \rho_{lm}(r) Y_{lm}(\theta, \phi) e^{-im\omega t} \\ &= \sum_{l=0}^{\infty} \left(\rho_{l0}(r) Y_{l0}(\theta, \phi) + \sum_{m=1}^l \left(\rho_{lm}(r) Y_{lm}(\theta, \phi) e^{-im\omega t} + \rho_{l,-m}(r) Y_{l,-m}(\theta, \phi) e^{im\omega t} \right) \right) \end{aligned}$$

Since $Y_{l,-m}(\theta, \phi) = (-1)^m Y_{lm}^*(\theta, \phi)$, and it is straightforward to check that $\rho_{l,-m}(r) = (-1)^m \rho_{lm}^*(r)$,

the charge distribution will only contain positive frequency components in the following form,

$$\rho(\vec{x}, t) = \sum_{l=0}^{\infty} \left(\rho_{l0} Y_{l0}(\theta, \phi) + \sum_{m=1}^l 2 \operatorname{Re} \left[\rho_{lm}(r) Y_{lm}(\theta, \phi) e^{-im\omega t} \right] \right)$$

Which can then be used for radiation calculations, with independent contributions for each individual frequency. The multipole moment can also be calculated from this expression.

The $m=0$ component is trivial,

$$q_{l0}(t) = \int d^3x \rho(\vec{x}, t) r^l Y_{l0}^*(\theta, \phi) = \int_0^{+\infty} r^l \rho_{l0} \cdot r^2 dr = q_{l0}.$$

and, for $m > 0$

$$\begin{aligned} q_{lm}(t) &= \int d^3x \rho(\vec{x}, t) r^l Y_{lm}^*(\theta, \phi) \\ &= \int_0^{+\infty} r^l dr \int d\Omega \sum_{m=1}^l \left(\rho_{lm}(r) Y_{lm}(\theta, \phi) e^{-im\omega t} + \rho_{l,-m}(r) Y_{l,-m}(\theta, \phi) e^{im\omega t} \right) Y_{lm}^*(\theta, \phi) r^2 \\ &= \int_0^{+\infty} r^{l+2} dr \left(\rho_{lm}(r) + \rho_{lm}(r) \right) e^{-im\omega t} = 2 q_{lm} e^{-im\omega t} \end{aligned}$$

where $q_{lm} = \int_0^{+\infty} r^{l+2} dr \int d\Omega r^2 \rho(\vec{x}) Y_{lm}^*(\theta, \phi).$

(b) The Fourier series for $\rho(\vec{x}, t)$ is $\rho(\vec{x}, t) = \sum_{n=-\infty}^{\infty} \rho_n(\vec{x}) e^{-in\omega_0 t}$, where

$$\rho_n(\vec{x}) = \frac{1}{T} \int_0^T \rho(\vec{x}, t) e^{-in\omega_0 t} dt. \text{ Clearly, } \rho_{-n}(\vec{x}) = \rho_n^*(\vec{x}). \text{ Then,}$$

$$\begin{aligned} \rho(\vec{x}, t) &= \rho_0(\vec{x}) + \sum_{n=1}^{\infty} (\rho_n(\vec{x}) e^{-in\omega_0 t} + \rho_{-n}(\vec{x}) e^{in\omega_0 t}) = \rho_0(\vec{x}) + \sum_{n=1}^{\infty} (\rho_n(\vec{x}) e^{-in\omega_0 t} + \rho_n^*(\vec{x}) e^{in\omega_0 t}) \\ &= \rho_0(\vec{x}) + \sum_{n=1}^{\infty} 2 \operatorname{Re} [\rho_n(\vec{x}) e^{-in\omega_0 t}] \end{aligned}$$

(c) Here, the charge density is $\rho(\vec{x}, t) = \frac{q}{r^2} \delta(r-R) \delta(\omega\sin\theta) \delta(\phi - \omega_0 t)$. Using the result from part (a), we have

$$\rho_{lm} = \int_0^{+\infty} r^2 dr \int d\Omega r^2 \rho(\vec{x}) Y_{lm}^*(0, \phi) = q R^2 Y_{lm}^*\left(\frac{\pi}{2}, 0\right) = q R^2 \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(0),$$

where $P_l^m(x)$ is associated Legendre polynomial. For $l=0$ and first few l and m , we know

$$P_0^0(0) = 1, \quad P_1^1(0) = -1, \quad P_2^0(0) = -\frac{1}{2}, \quad P_2^2(0) = 3 \text{ and zero for other combinations with } l \leq 2.$$

Therefore, $\rho_{00} = \frac{q}{\sqrt{4\pi}}, \quad \rho_{11} = -qR \sqrt{\frac{3}{8\pi}}, \quad \rho_{20} = -\frac{qR^2}{2} \sqrt{\frac{5}{4\pi}}, \quad \rho_{22} = 3qR^2 \sqrt{\frac{5}{96\pi}}$

From part (b),

$$\rho_n(\vec{x}) = \frac{q}{r^2} \delta(r-R) \delta(\omega\sin\theta) \delta(\phi).$$